A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 7: Ethics and Ownership

Subtopic 7.1: Ethics and Ownership

Concept 7.1.4: Artificial Intelligence (Impact and Applications)

Total Questions: 13

Mark Schemes begin on Page 9

Question 1 | Source: 9618_w25_qp_11 (Q8.c)

8 One ethical consideration for a student connecting their personal computer to the school network is the risk of spreading malware on the network.

(c) Describe **two** social impacts of students using Artificial Intelligence (AI) to complete their homework.

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[4]

Question 3 | Source: 9618_s25_qp_11 (Q4.a)

4 A shop installs a new system that allows users to purchase items without going through a manual checkout.

The new system:

- identifies customers when they enter the shop and matches them to their account
- prevents a customer from walking through the automatic barriers if they do not have an account
- automatically detects the items that a customer has taken from a shelf and charges these to the customer's account.

(a) The new system uses digital cameras and Artificial Intelligence (AI) to identify the customers.

Explain how the new system uses AI to identify each customer.

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Question 4 | Source: 9618_s25_qp_12 (Q3.a)

3 A computer program uses a digital camera to read the words on an item.

The program can read the words that are written on the item, translate the words to a chosen language and then output the words as audio. For example, when used in a supermarket, the program can output the words written on the labels on products.

The program uses Artificial Intelligence (AI).

(a) Explain how AI is used in the computer program described.

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Question 5 | Source: 9618_s25_qp_12 (Q3.b)

3 A computer program uses a digital camera to read the words on an item.

The program can read the words that are written on the item, translate the words to a chosen language and then output the words as audio. For example, when used in a supermarket, the program can output the words written on the labels on products.

The program uses Artificial Intelligence (AI).

(b) State **two** social benefits of the use of AI in the computer program described.

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Question 6 | Source: 9618_w24_qp_11 (Q6)

6 A car park system uses a camera to record the registration number of each car as it enters and leaves the car park.

Explain how artificial intelligence is used in the car park system to identify the car's registration number.

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Question 7 | Source: 9618_s24_qp_11 (Q5.c.ii)

5 A bank allows customers to access their accounts using an application that they can download onto a device such as a smartphone.

(c) The bank also needs to keep its customers' data private and secure.

(ii) Customers need to use biometric authentication to access their accounts.
One biometric authentication method is facial recognition.

Facial recognition uses Artificial Intelligence (AI).

Describe how AI is used in facial recognition.

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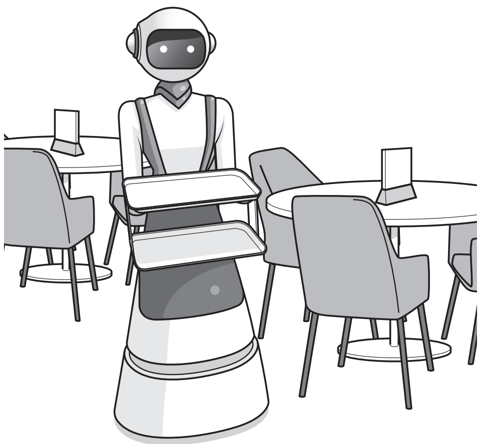
Question 8 | Source: 9618_s24_qp_13 (Q7.b)

7 Robots are used to serve food and drink to customers at a restaurant.



(b) The robot uses Artificial Intelligence (AI) to communicate with the customers. The customers speak to the robot to order their food and drinks.

Explain how AI will be used in this part of the robot.

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Question 9 | Source: 9618_s23_qp_11 (Q4.c)

4 A networked closed-circuit television (CCTV) system in a house uses sensors and cameras to detect the presence of a person. It then tracks the person and records a video of their movements.

Data from the CCTV cameras is transmitted to a central computer.

(c) The CCTV system uses Artificial Intelligence (AI) to identify the presence of a person in the house and to track their movements.

Describe how AI is used in this system.

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Question 10 | Source: 9618_s23_qp_12 (Q7.d)

7 A software developer is working in a team writing a program for a client.

(d) One section of the program being developed will convert user's speech into commands.

Explain how Artificial Intelligence (AI) can be used in this program.

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Question 11 | Source: 9618_s23_qp_13 (Q3.b)

3 A mobile telephone is used to record a video.

(b) The mobile telephone uses a built-in digital camera to record the video.

The digital camera automatically focuses on the faces of people.

Explain how Artificial Intelligence (AI) is used by the camera to automatically focus on the faces of people.

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Question 12 | Source: 9618_w22_qp_12 (Q9)

9 One use of Artificial Intelligence (AI) is for facial recognition software.

Describe the social impact of using facial recognition software to identify individuals in an airport.

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Question 13 | Source: 9618_w22_qp_13 (Q8.b)

(b) A commercial program for a vehicle repair garage includes an Artificial Intelligence (AI) module that can diagnose faults and suggest repairs.

Describe **one** economic impact the AI module may have on the garage.

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Question 14 | Source: 9618_s21_qp_12 (Q2.b)

2 Aisha manages a team of software developers.

(b) The team are developing a computer game where the user plays a board game (such as chess) against the computer.

Describe how the computer would use Artificial Intelligence (AI) to play the board game.

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MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w25_ms_11 (Q8.c)

8(c)	**1 mark** per bullet point, **max 4** marks e. g. **Negative Impacts** • If students use AI to shortcut learning, they may miss out on developing reasoning and problem-solving skills // Students may not be learning anything // It could lead to a decline in specific skills • It could have negative impacts on student communication // heavy reliance on AI may limit opportunities for collaboration, teamwork, and face-to-face communication • Emphasises the digital divide// some students may not have access to the technology • The AI answer might not be correct leading to mis-information • Increased screen time and isolation could contribute to anxiety or loneliness **Positive Impacts** • Students given additional support may do better • Pupils who struggle in traditional settings may gain self-esteem through AI-assisted learning at their own pace **Long-Term Impacts** • Protection of intellectual property // ethical issues and considerations • Schools may need to redesign curricula and/or assessment to integrate AI	**4**

Mark Scheme for Question 2 | Source: 9618_w25_ms_13 (Q7.a)

7(a)	**1 mark** per point, **max 2** marks e.g. • The learning experience could be improved • … by identifying students who are struggling • …. and employing more personalised learning • There could be early Intervention • … to determine which students require extra support • • Teachers and students could understand the challenges better • …and identify optimal times for learning • There could be privacy concerns • …. students / teachers could be uncomfortable with constant surveillance • ... students might not want their data / actions passed to third parties • There could be Mental Health concerns • …. due to constant monitoring and pressure • …. contributing to student stress and anxiety	**2**

Mark Scheme for Question 3 | Source: 9618_s25_ms_11 (Q4.a)

4(a)	**1 mark** each to **max 4** e.g. • Uses image recognition / facial recognition • Measures the distance between facial features of the customer • and stores the values in a database of user information • An image is captured using the digital camera • the AI identifies that the image is a face • by analysing the pixels to find patterns • the distance between features is calculated and compared to those in database	**4**

Mark Scheme for Question 4 | Source: 9618_s25_ms_12 (Q3.a)

3(a)	**1 mark** each to **max 4** e.g. • Uses image recognition // optical character recognition • to analyse a photograph/pixels to identify the location of characters within an image • in order to convert the pattern of pixels into individual characters • Uses natural language • to combine the characters into words • and compare the words to database to identify a match in the language of the writing • it may also predict what is missing / incomplete / from what has been read • then compare the words to a database of the user's chosen language to find a match and translate • Uses text-to-speech • to generate the audio waveforms corresponding to the words	**4**

Mark Scheme for Question 5 | Source: 9618_s25_ms_12 (Q3.b)

3(b)	**1 mark** each to **max 2** e.g. • It can help people with visual impairment to identify which products to purchase • It can help people who cannot understand the language • It can help to overcome learning / reading difficulties	**2**

Mark Scheme for Question 6 | Source: 9618_w24_ms_11 (Q6)

6	**1 mark** for each bullet point (**max 4**) e.g. • It uses image recognition • The pixels of each image from the camera are stored • … and matched to the expected shape/size/colour of registration number • It uses optical character recognition • The pixels within the registration number area identified are analysed • … and compared to expected characters/letters/numbers	**4**

Mark Scheme for Question 7 | Source: 9618_s24_ms_11 (Q5.c.ii)

5(c)(ii)	**1 mark** each to **max 4**: e.g. • Captures an image of the face • Uses image recognition • Trained to identify the features of a face in an image • … using a large number of images • Analyse images for facial features • Uses the probability of a match	**4**

Mark Scheme for Question 8 | Source: 9618_s24_ms_13 (Q7.b)

7(b)	**1 mark** each to **max 3**: e.g. • Voice/speech recognition is used • ... to identify if someone speaking • The sound is recorded and analysed • The audio recordings are compared to a database of words/sound waves • ... to identify the word that has the highest probability of being said • Natural language recognition is used • Words are combined and compared to known sentences • … programmed action(s) for matching sentence(s) are performed	**3**

Mark Scheme for Question 9 | Source: 9618_s23_ms_11 (Q4.c)

4(c)	**1 mark** each to **max 3** Examples: • Uses image recognition • Monitors every image taken to identify matching images/shapes/features to a 'person' … • … starts recording to secondary storage/permanently when a person is identified • System identifies direction of movement of person and uses this to decide where/how to move the camera/record • System identifies other cameras to start recording based on direction of movement	**3**

Mark Scheme for Question 10 | Source: 9618_s23_ms_12 (Q7.d)

7(d)	**1 mark** each to **max 3** • Uses speech recognition • ... which identifies key phrases / words spoken • … and matches these to a database • ... and generates the most likely sentence / command / word	**3**

Mark Scheme for Question 11 | Source: 9618_s23_ms_13 (Q3.b)

3(b)	**1 mark** each to **max 3** Examples: • Scans the scene in real time • Identifies if there are faces in the image • Uses facial recognition • ... uses image recognition • ... takes each frame individually • ... analyses the pixels • ... stores pattern for a face • ... looks for patterns that match/come close to the pattern for a face • Camera focuses on the **pattern** identified	**3**

Mark Scheme for Question 12 | Source: 9618_w22_ms_12 (Q9)

9	**1 mark** for each bullet point **(max 2)**. For example: • incorrect recognition of faces leads to mistakes such as • ... access to facilities / systems may be denied • privacy issues / people do not like data being stored • individuals will feel safer • ... there might be a reduction in crime • faster boarding • catching criminals	**2**

Mark Scheme for Question 13 | Source: 9618_w22_ms_13 (Q8.b)

8(b)	**1 mark** for a correct economic impact and **1 mark** for corresponding description Example: • reduce costs to the garage • ... because less time taken for diagnosis • increase profits for the garage • ... as technicians spend more time repairing, so completing more jobs in a day • decrease costs passed to customer • ... so garage may gain customers • profit margins can be reduced • ... because program may be expensive to buy / maintain / update	**2**

Mark Scheme for Question 14 | Source: 9618_s21_ms_12 (Q2.b)

2(b)	**1 mark** per bullet point to **max 3** • The rules / past moves / decision making algorithms of the game will be stored • The AI program is trained, by playing many times • AI will look (ahead) at possible moves • ... and/or analyse the pattern of past choices • … and choose the move most likely to be successful • Computer could learn how to improve // learn from previous mistakes • … by storing the positive/negative result of choices • ... and changing its future choices	**3**